

4. Payment Flow

This section describes the end-to-end lifecycle of a card payment in the system, from the moment a customer initiates a payment to the final transfer of funds to the merchant.

4.1 High-Level Payment Flow

The typical flow of a card-based payment system is as follows:

1. The **merchant's customer** initiates a payment using a card.
2. The **merchant** creates a **Payment Intent** in Stripe (payment request).
3. Stripe processes the request and communicates with the **Card Network**.
4. The Card Network routes the transaction to the **Issuing Bank** for approval.
5. Once approved, funds move from the customer's bank to Stripe's **Acquiring Bank**.
6. Stripe records the transaction in its **internal ledger** and deducts a processing fee.
7. Stripe later initiates a **payout** to the merchant's bank account.

4.2 Role of Card Networks

Since we are dealing with card payments (not direct bank transfers), Stripe does not directly interact with bank accounts.

Instead:

- Stripe sends transaction requests to **Card Networks (Visa, Mastercard, etc.)**
- Card Networks act as intermediaries between:
 - Issuing Bank (customer's bank)
 - Acquiring Bank (Stripe's bank)

4.3 Money Movement Lifecycle

The actual movement of money happens in three distinct stages:

1. Authorization

- The issuing bank verifies:
 - Card validity
 - Available balance

- Fraud risk signals
- If approved, the bank **reserves (locks)** the required amount
- No actual money transfer happens yet

Think of this as:

- SQL analogy: **begin transaction**
- Money is “reserved”, not transferred

2. Capture

- Stripe sends a **capture request**
- This is an **idempotent operation** (safe to retry without duplication)
- It confirms that the authorized amount should be transferred
- The reserved funds are now marked for transfer

Think of this as:

- SQL analogy: **commit transaction**
- Money movement is officially confirmed

3. Settlement

- Final step where actual money moves between banks:
 - Issuing bank → Acquiring bank
- This may happen in batches (not instant)
- Can take hours or days depending on banking rails

Think of this as:

- Database analogy: **replication to follower nodes**
- Final consistency across financial systems

4.4 Internal Stripe Processing

Once payment is successful:

- Stripe records the transaction in its **Ledger system**
- Ledger ensures:
 - Debit from customer-side account
 - Credit to merchant-side account

- Stripe deducts its **transaction fee**
- Balance updates are always consistent and auditable

4.5 Conceptual Analogy

We can map payment flow to database concepts:

Payment Stage	Database Analogy	Meaning
Authorization	BEGIN TRANSACTION	Reserve funds
Capture	COMMIT TRANSACTION	Confirm payment
Settlement	Replication to followers	Final money movement

4.6 Key Takeaways

- Payment flow is **multi-stage and asynchronous**
- Authorization $\neq$ actual money transfer
- Capture is the **commit point**
- Settlement is **final reconciliation between banks**
- Ledger ensures internal financial correctness at Stripe level